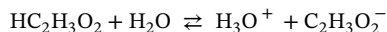


The equilibrium



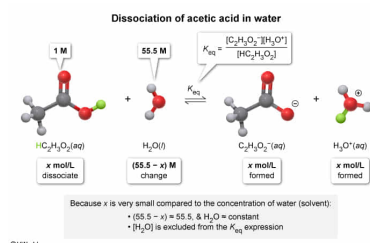
exists in a 1 M aqueous solution of  $\text{HC}_2\text{H}_3\text{O}_2$  with an equilibrium constant  $K_{\text{eq}} = [\text{H}_3\text{O}^+][\text{C}_2\text{H}_3\text{O}_2^-]/[\text{HC}_2\text{H}_3\text{O}_2]$ . Which of the following expressions explains why  $K_{\text{eq}}$  does not include a term for  $[\text{H}_2\text{O}]$ ?

- ☐ A.  $[\text{H}_2\text{O}] < [\text{C}_2\text{H}_3\text{O}_2^-]$  [2%]
- ✓ ☒ B.  $[\text{H}_2\text{O}] \approx 55.5 \approx \text{constant}$  [56%]
- ☐ C.  $[\text{H}_3\text{O}^+] > [\text{H}_2\text{O}]$  [5%]
- ☐ D.  $[\text{H}_2\text{O}] \approx 0$  [35%]

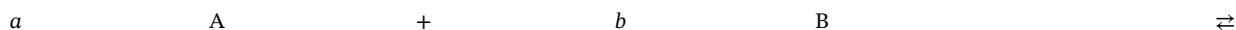
Correct

56% answered correctly

Explanation:



The general reaction in which chemical species A and B react to form products C and D is written as



where  $a$ ,  $b$ ,  $c$ , and  $d$  are the coefficients of the balanced reaction. For a reaction at equilibrium, the **law of mass action** states that an **equilibrium constant**  $K_{\text{eq}}$  can be expressed as a **ratio** of the molar concentrations of the products over the reactants, each raised to the power of its respective balanced **reaction coefficient**:

$$K_{\text{eq}} = \frac{\left[ \frac{\text{products}}{\text{coefficients}} \right]}{\left[ \frac{\text{reactants}}{\text{coefficients}} \right]}$$

However, **solids, pure liquids, and solvent** species are **omitted** from the  $K_{\text{eq}}$  **expression** because the activities of these species are not approximated by their molar concentrations. For example, a 1 M aqueous solution of  $\text{HC}_2\text{H}_3\text{O}_2$  with an equilibrium reaction of



has an associated  $K_{\text{eq}}$  expression of

$$K_{\text{eq}} = \frac{[\text{C}_2\text{H}_3\text{O}_2^-][\text{H}_3\text{O}^+]}{[\text{HC}_2\text{H}_3\text{O}_2]}$$

In this reaction, the concentration of water is omitted from the  $K_{\text{eq}}$  expression because, although water is a reactant, it is also the solvent. As the solvent, the number of water molecules is significantly greater than the number of all other species in the solution; the  $\text{HC}_2\text{H}_3\text{O}_2$  is dilute and dissociates only to a small extent. Therefore, in relative terms, the  $[\text{H}_2\text{O}]$  (55.5 M) is essentially constant and is omitted.

**(Choices A and C)** Water is the solvent and the  $\text{HC}_2\text{H}_3\text{O}_2$  concentration is low; therefore, the number of water molecules per unit volume is *vastly greater* than the number of molecules of  $\text{HC}_2\text{H}_3\text{O}_2$  or the number of  $\text{H}_3\text{O}^+$  or  $\text{C}_2\text{H}_3\text{O}_2^-$  ions formed as dissociation products.

**(Choice D)** As the solvent, water cannot have a concentration of zero.

**Educational objective:**

Applying the law of mass action, an equilibrium constant  $K_{\text{eq}}$  can be expressed for a reaction as a ratio of the molar concentrations of the products over the reactants, each raised to the power of its respective balanced reaction coefficient. Because the concentration of the solvent is much greater than the concentration of all other species in the solution, the solvent is relatively constant and is omitted from the  $K_{\text{eq}}$  expression.

Time spent: 0 sec  
Subject:  
General Chemistry

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System:  
Thermodynamics, Kinetics & Gas Laws